FIREWATCH SA — Final KDE Basemap Handover

Final group-selected design | Version 5.0 | 23 October 2026

1. What was selected

The group selected the FIREWATCH SA bright-theme, seven-tab R Shiny dashboard developed with the original NASA FIRMS VIIRS NOAA-21 (2024–2025) dataset. The selected enhancement is an interactive OpenStreetMap basemap beneath the Kernel Density Estimation (KDE) raster. This is the user's design, not the alternate group member's application.

2. Project files

- app.R — final seven-tab script with interactive KDE basemap
- README.md — setup, folder structure, deployment and limitations
- 01_Project_and_Script_Guide — dashboard architecture and statistical explanation
- 02_Datasets_and_Sources — sources, links and dataset inventory
- 03_Meeting_and_Presentation_Plan — roles, demonstration, questions and contingency
- 04_Final_KDE_Basemap_Handover — this update and testing checklist
- source_archives/ — source data archives, for provenance; not the prepared RDS files
- data/ — copy fires_final.rds and provinces.rds from the existing working project

3. How the KDE basemap works

User inputs: The user selects the province or All Provinces, year, season, and KDE bandwidth (2–30 km), then clicks Run KDE Analysis.

Analysis window and projection: The app transforms fire detections and boundaries into a projected coordinate system in meters: local UTM for one province or South Africa-centered Lambert azimuthal equal-area for national analysis.

Point sampling: Up to 6,000 detections are sampled reproducibly (seed 123) for a manageable spatial calculation. The raster is rescaled by the filtered-record/sample ratio to estimate detection intensity.

KDE calculation: spatstat.explore::density.ppp computes the smoothed intensity surface on a 180 × 180 pixel grid. The bandwidth controls the smoothing scale.

Georeferencing: The pixel grid is converted into a terra raster, its rows are oriented north-up, and it is projected to WGS84 (EPSG:4326).

Interactive map: Leaflet draws OpenStreetMap tiles, then the transparent KDE raster, province boundaries, an intensity legend, and pan/zoom controls.

Export: The existing PNG download draws the statistical KDE surface as a static figure; it does not export online map tiles.

4. How to interpret the KDE map

Bright or high-value areas represent places where the selected satellite detections are concentrated after smoothing. They do not directly indicate burned area, fire severity, unique wildfire incidents, or statistically significant hotspots. The map depends on the chosen bandwidth and sampling. Basemap labels help locate the patterns geographically; they are not explanatory variables in the model.

5. Other retained analyses

Nearest Neighbour compares observed average distances with a uniform spatial-randomness benchmark and 99 Monte Carlo simulations. Ripley's K is presented as L(r) − r with 19 pointwise simulation envelopes. The seasonal tab compares 2024 and 2025 detections and province/season patterns. Interpretation panels update from the computed results.

6. Dataset sources and attribution

- **NASA FIRMS:** https://firms.modaps.eosdis.nasa.gov/
- **FIRMS downloads:** https://firms.modaps.eosdis.nasa.gov/download/
- **FIRMS data guide:** https://firms.modaps.eosdis.nasa.gov/download/Readme.txt
- **OpenStreetMap attribution:** https://www.openstreetmap.org/copyright

The exact source/version of the prepared provincial boundaries has not been verified. Ask the original data preparer before assigning a source citation. No data from another team member's satellite product were merged.

7. Installation and reproducibility

Install: shiny, sf, dplyr, ggplot2, leaflet, scales, spatstat.geom, spatstat.explore, spatstat.random, terra. Add data/fires_final.rds and data/provinces.rds. The two RDS files are required to run the dashboard and are not included in the ZIP.

8. Group acceptance checklist

- Copy both original RDS files into data/
- Open the updated app.R in RStudio and click Run App
- Verify seven tabs and the bright theme
- Run KDE for a province: basemap, colored overlay, legend and boundaries visible
- Pan/zoom and change bandwidth, then rerun KDE
- Run All Provinces KDE and one NN/Ripley's K test
- Confirm dynamic interpretation panels and CSV/PNG downloads
- Deploy only after successful local testing; verify the live app in an incognito window
- Rehearse an offline/static screenshot fallback if the server disconnects

9. Validation status

The source code was modified and checked for the required KDE-basemap elements, but it could not be executed end-to-end here because the prepared RDS files and an R runtime were not available. The group must run the acceptance checklist before calling this the final working deployed build. The previously shared live URL may still host an older build.